StockCL: Selective Contrastive Learning for Stock Trend Forecasting via Learnable Concepts

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Abstract. Stock trend forecasting is crucial for quantitative investment and various deep learning models have been proposed to obtain superior performance. Due to the limited stock data, existing deep learning models suffer from overfitting. It has been revealed that supervised contrastive learning can provide additional supervision signals by pulling closer samples with the same label and pushing apart samples with different labels. However, stock data has continuous labels and it is nontrivial to identify appropriate contrastive pairs based on label information. In this paper, we develop a novel selective contrastive learning framework named StockCL for stock trend forecasting, which is applicable to any stock trend forecasting models. Our key insight is to identify latent concepts that drive the stock trends and select reliable contrastive pairs according to the samples' belonging concepts and their label similarity. Experiments on two datasets with four stock trend forecasting models demonstrate that StockCL consistently improves the forecasting performance with a significant margin by 5%-18%. Code is available at https://github.com/Starr8845/StockCL.git.

Keywords: Stock trend forecasting · Contrastive learning

1 Introduction

Stock trend forecasting, essential for precise investment decisions and portfolio optimization, involves predicting daily stock change rates based on historical movement features. However, a persistent challenge of deep learning on this task is the data-hungry issue [14], where limited data leads to easy overfitting of deep learning models, hampering their generalization and backtest performance. It is thus crucial to improve the training paradigm and mitigate the data hungry and overfitting issue with more supervision signals.

Recent studies demonstrate the efficacy of contrastive learning in enhancing the generalization of deep learning models by introducing additional supervision signals. It generates positive and negative pairs, pulling representations of positives closer and pushing negatives further to learn robust representations. Contrastive learning can be categorized into self-supervised and supervised methods based on label usage. Self-supervised approaches form positive pairs through

data augmentations. However, designing augmentations for stock trend data is challenging, and some methods like adding Gaussian noise [3] can distort underlying patterns. Supervised contrastive learning [9, 8] constructs positive pairs based on labels without augmentations. With continuous labels, contrastive regression methods in tasks like gaze estimation [16] and medical image analysis [1] choose positive pairs based on label proximity, assuming samples with similar labels should have akin representations.

Though supervised contrastive learning based on label similarity can be applied to stock data, we argue that the generated contrastive pairs are suboptimal. Due to the complexity of stock markets, samples with similar labels (*i.e.*, price change rate) can exhibit distinct input features (*i.e.*, historical trends). For example, two stocks might share similar increases, but one may be experiencing upward momentum while the other is in a prolonged decline. Treating such samples as positive pairs introduces numerous false positives, misleading the training process and harming predictive performance. Thus, a more selective criterion towards reliable contrastive pairs in stock data is urgently needed.

It is well acknowledged that stock trends are driven by latent concepts [18] such as market sentiment, capital flows, and broader market conditions, etc. Samples sharing the same concept often exhibit correlated trends. For instance, in the concept of ongoing favorable conditions, stock prices typically trend upwards alongside sustained high trading volumes. Motivated by this, our key insight is that samples sharing the same concept should have closer representations. We identify latent concepts based on sample input features and use them to filter noisy positive pairs generated solely by label similarity. Our approach constructs positive pairs by incorporating both label and concept information.

In this paper, we propose a novel selective supervised contrastive learning framework named StockCL for stock trend forecasting via learnable concepts. StockCL comprises two main components: the concept pool and the concept-aware contrastive pair selector. The concept pool learns concept representations and assigns each sample a belonging concept. The concept-aware contrastive pair selector utilizes both concept and label information to choose reliable contrastive pairs and trains sample representations through an extended contrastive loss. The parameters are jointly optimized through the two components' losses and supervised training in an end-to-end fashion.

The main contributions of this paper can be summarized as:

- We propose StockCL, a novel and model-agnostic supervised contrastive learning framework for stock trend forecasting to mitigate data-hungry and overfitting issues.
- We develop the concept pool and concept-aware contrastive pair selector components in StockCL. The former learns concept representations in the embedding space, and the latter constructs reliable contrastive pairs by considering sample concepts and label similarity.
- We conduct extensive experiments on two real-world datasets and four stock trend forecasting models. StockCL consistently improves the models' stock trend forecasting performance from 5% to 18%.

2 Preliminaries

In this section, we define the problem of stock trend forecasting and give the key notations used throughout the paper.

Stock features. At time step t, the features of a stock s can be denoted as $\mathbf{x}_t^{(s)} \in \mathbb{R}^D$, where D is the feature dimension.

Stock trend. At time step t, the stock trend (label) of stock s is defined as the relative change rate of the stock price: $y_t^{(s)} = \frac{Price_{t+1}^{(s)} - Price_t^{(s)}}{Price_t^{(s)}} \in \mathbb{R}$.

Stock trend forecasting. Stock trend forecasting is to learn a forecasting model $\mathcal{F}(\cdot)$ to predict $y_t^{(s)}$ given a collection of daily features on the past time window $\mathbf{X}_t^{(s)} = [\mathbf{x}_{t-T+1}^{(s)}, ..., \mathbf{x}_{t-1}^{(s)}, \mathbf{x}_t^{(s)}] \in \mathbb{R}^{T \times D}$. From a representation learning point of view, the forecasting model $\mathcal{F}(\cdot)$ can be split into an encoder $f(\cdot)$ and a predictor $p(\cdot)$. The encoder $f(\cdot)$ takes $\mathbf{X}_t^{(s)}$ as input, acts as a feature and semantics extractor and outputs a d dimension representation $\mathbf{z}_i$ in the embedding space and the predictor $p(\cdot)$ takes the representation as input and gives the final prediction, $\mathbf{z}_t^{(s)} = f(\mathbf{X}_t^{(s)}) \in \mathbb{R}^d,\ \hat{y}_t^{(s)} = p(\mathbf{z}_t^{(s)})$.

Training samples. The training samples include $\{(\mathbf{X}_t^{(s)}, y_t^{(s)})\}$ across all stocks and all time steps. We denote all the training samples as $\mathcal{T} = \{(\mathbf{X}_i, y_i)\}_{i=1}^N$, where N is the total number of samples.

3 The StockCL Framework

Overview. As shown in Fig. 1, the proposed StockCL framework comprises two core components: (i) The concept pool learns a set of concept representations in the embedding space using a concept loss and matches each sample with its belonging concept; (ii) The concept-aware contrastive pair selector selects reliable contrastive pairs based on both concept and label information and trains the representations by an extension of InfoNCE [11] loss.

3.1 Concept pool

To group samples dominated by the same market style, we employ a set of learnable concept representations in the concept pool and design a concept loss to train the concept representations.

Concept representations. First, we randomly initialize M concept representations in the embedding space, of which each represents a typical market style. The concept representations are denoted as $\mathbf{C} = \{\mathbf{c}_1, \mathbf{c}_2, ..., \mathbf{c}_M\}$, where $\mathbf{c}_k \in \mathbb{R}^d$ is the kth concept representation.

Concept matching. We compare each training sample with all the concepts to retrieve the concept it belongs to. For the training sample with representation $\mathbf{z}_i$, the matching score between the sample and the the kth concept is

$$\alpha_{ik} = \frac{\exp\left(\mathbf{z}_i^\top \mathbf{c}_k\right)}{\sum_{m=1}^{M} \exp\left(\mathbf{z}_i^\top \mathbf{c}_m\right)}. \tag{1}$$

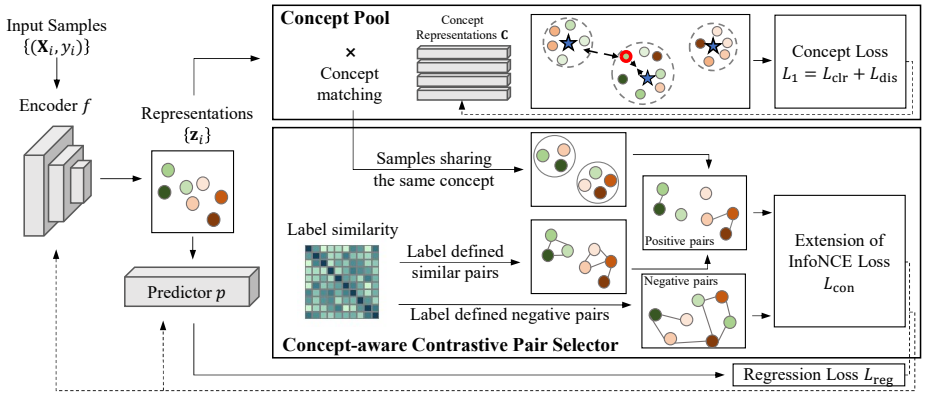

Fig. 1. Overview of StockCL framework.

We take the concept with the highest matching score as the retrieved concept for $\mathbf{z}_i$, and the retrieved concept index is formulated as $d_i = \arg\max_k \alpha_{ik}$.

Concept loss. Given the sample representations, we optimize the concept representations in the embedding space. Inspired by works in memory networks [7] that focus on memorizing typical features for future pattern matching, we introduce a concept loss to train the concept representations. Ideally, each learned concept should be in the cluster center of the samples sharing this concept. At the same time, the learned concepts should be discriminative on the samples to make sure that each sample's matched concept is the dominant one. Hence, the concept loss contains two parts, denoted as L_{clr} and L_{dis}.

The first part L_{clr} measures the distance between the concept and the samples sharing this concept in the embedding space. We minimize the distance in the form of MSE loss to make sure each learned concept is in the corresponding cluster center:

$$L_{\text{clr}} = \sum_{k=1}^{M} \sum_{i \in \mathcal{T}} \mathbb{I}(d_i = k) \|\mathbf{c}_k - \mathbf{z}_i\|_2^2, \tag{2}$$

where $\mathbb{I}$ is the indicator function.

The second part L_{dis} measures whether the matched concept is the dominant one for each sample. L_{dis} is in the form of triplet loss [13], which computes the distance gap between the sample representation (anchor) and the nearest concept (positive) and the second nearest concept (negative). With a hyper-parameter $\delta > 0$ as a lowest margin between the two distances, the nearest concept will be pulled closer to the sample and the second nearest concept will be pushed apart until the gap is greater than δ. L_{dis} can be denoted as

$$L_{\text{dis}} = \sum_{i \in \mathcal{T}} \max\left(0, \|\mathbf{z}_i - \mathbf{c}_{d_i}\|_2^2 - \|\mathbf{z}_i - \mathbf{c}_{d_i^{(2)}}\|_2^2 + \delta\right), \tag{3}$$

where $d_i^{(2)} = \arg\max_{k,k\neq d_i} \alpha_{ik}$ is the the concept index with the second highest matching score for $\mathbf{z}_i$.

We take the sum of the two parts as the concept loss, $L_1 = L_{\text{com}} + L_{\text{dis}}$, to train the concept representations in the embedding space. L_1 is back-propagated only on the concept representations $\mathbf{C}$, without gradients to the encoder and the predictor.

3.2 Concept-aware contrastive pair selector

In concept-aware contrastive pair selector component, we select reliable contrastive pairs based on both concept and label information and use them in an extension of InfoNCE [11] loss to train the sample representations.

Contrastive pair selection. With the retrieved concept for each sample in the concept pool, we define positive pairs as those with similar labels and sharing the same concept. In this way, we can filter the noisy positive pairs generated by label similarity. For sample $(\mathbf{X}_i, y_i)$, we define the positive sample set $\mathcal{S}_i^{pos}$ of this anchor sample based on its label y_i and concept index d_i,

$$\mathcal{S}_i^{pos} = \{(\mathbf{X}_j, y_j) \in \mathcal{A} | d_i = d_j, \mathcal{d}(y_i, y_j) < \epsilon^{pos}\}, \tag{4}$$

where $\mathcal{d}(\cdot,\cdot)$ is a pre-defined label distance function and we simply instantiate it as $\mathcal{d}(a,b) = |a-b|$. ϵ^{pos} is the positive pair threshold on label similarity and $\mathcal{A}$ represents the training batch of samples.

At the same time, we want to explicitly enlarge the distances between the anchor sample and the samples with dissimilar labels. Hence, rather than taking the whole batch $\mathcal{A}$ as negative samples, we define the negative sample set of each anchor sample, by label similarity,

$$\mathcal{S}_i^{neg} = \{(\mathbf{X}_j, y_j) \in \mathcal{A} | \mathcal{d}(y_i, y_j) > \epsilon^{neg}\}, \tag{5}$$

where ϵ^{neg} is the negative threshold on label similarity and $\epsilon^{neg} > \epsilon^{pos}$.

Extension of InfoNCE. With the selected positive and negative sample sets for each anchor, we employ contrastive learning to enhance representations. We extend the infoNCE [11] loss to accommodate label differences of each pair and ensure tighter alignment between samples sharing more close labels. Pair weight β_{ij} for positive pair $\mathbf{z}_i$ and $\mathbf{z}_j$ is determined by their concept similarity,

$$\beta_{ij} = \boldsymbol{\alpha}_i^\top \boldsymbol{\alpha}_j, \quad \boldsymbol{\alpha}_i = [\alpha_{i1}, \alpha_{i2}, ..., \alpha_{iM}]^\top. \tag{6}$$

With pair weights, the extension of InfoNCE loss can be denoted as follows:

$$L_{\text{con}} = \sum_{i\in\mathcal{T}} L_{\text{con}}^i = \sum_{i\in\mathcal{T}} -\frac{1}{|\mathcal{S}_i^{pos}|} \sum_{j\in\mathcal{S}_i^{pos}} \beta_{ij} \log \frac{\exp(\mathbf{z}_i^\top \mathbf{z}_j/\tau)}{\sum_{k\in\mathcal{S}_i^{neg}} \exp(\mathbf{z}_i^\top \mathbf{z}_k/\tau)}. \tag{7}$$

where τ is the softmax temperature to control the uniformity.

Table 1. Experimental datasets

	Training			Validation			Test		
	#days	#stocks	#samples	#days	#stocks	#samples	#days	#stocks	#samples
CSI300	1684	505	478K	456	386	128K	868	451	253K
CSI500	1684	1058	785K	456	677	205K	868	810	418K

3.3 Overall training procedure of StockCL

The training loss of the encoder and predictor L_2 is taken as the sum of supervised regression loss and contrastive loss,

$$L_2 = L_{\text{reg}} + \lambda L_{\text{con}}, \quad L_{\text{reg}} = \sum_{i \in \mathcal{T}} (\hat{y}_i - y_i)^2, \tag{8}$$

where L_{reg} is the MSE between predictions and labels, and λ is a hyperparameter to control the strength of the contrastive loss.

In the overall training procedure, we first warm up the representations without contrastive loss and then train the parameters using all the losses. The concept representations within the concept pool are initialized randomly. Considering the poor quality of sample representations and concept representations in the early stage, we warm up the framework for a few epochs, using L_{reg} to train the the encoder and the predictor and L_1 to train the concept representations, without the noisy contrastive pairs. After the warm up stage, we train the concept representations, the encoder and the predictor using L_1 and L_2. All the parameters in the framework are jointly optimized in an end-to-end fashion.

4 Experiments

4.1 Experiment settings

Datasets. Following the Qlib benchmark[1], we test StockCL on two real world datasets, CSI300 and CSI500 in China A-share market to access the performance improvement. The statistics of the two datasets are shown in Tab. 1. We use the Alpha360 feature set in Qlib, which contains the opening price, the closing price, the highest price, the lowest price and the trading volume in each day and takes the features in the past 60 days as sample features.

Stock Forecasting Models. We employ three kinds of representative deep models for stock trend forecasting: (1) RNN-based methods, including GRU and ALSTM [12]; (2) Localformer [6] as a representative Transformer-based method; (3) GAT [15] as a representative GNN, which considers the impact of stock relationships on trends.

[1] https://github.com/microsoft/qlib

Table 2. Performance comparison between StockCL and baseline methods on CSI300 and CSI500 datasets, using different stock trend forecasting models.

Model	Method	CSI300				CSI500			
		IC	ICIR	RankIC	RankICIR	IC	ICIR	RankIC	RankICIR
GRU	-	0.055	0.453	0.056	0.474	0.053	0.584	0.054	0.603
	ConAlign	0.058	0.460	0.052	0.441	0.057	0.596	0.048	0.524
	CRGA	0.053	0.415	0.054	0.438	0.052	0.531	0.050	0.515
	RNC	0.059	0.473	0.053	0.451	0.057	**0.599**	0.052	0.567
	StockCL	**0.063**	**0.533**	**0.056**	**0.497**	**0.059**	0.583	**0.057**	**0.604**
ALSTM	-	0.053	0.365	0.053	0.362	0.052	0.533	0.056	0.620
	ConAlign	0.055	0.397	0.054	0.380	0.053	0.540	0.050	0.508
	CRGA	0.050	0.394	0.048	0.378	0.048	0.462	0.041	0.398
	RNC	0.058	0.471	0.056	0.474	0.053	0.526	0.055	0.599
	StockCL	**0.066**	**0.546**	**0.056**	**0.502**	**0.059**	**0.604**	**0.057**	**0.628**
Local-former	-	0.052	0.460	0.053	**0.488**	0.048	0.472	0.050	0.532
	ConAlign	0.053	0.432	0.048	0.406	0.050	0.519	**0.053**	**0.565**
	CRGA	0.033	0.220	0.028	0.178	0.027	0.261	0.029	0.273
	StockCL	**0.059**	**0.471**	**0.054**	0.455	**0.053**	**0.565**	0.048	0.557
GAT	-	0.054	0.434	0.047	0.398	0.048	0.479	0.048	0.485
	ConAlign	0.053	0.414	0.049	0.408	0.051	0.524	0.050	0.524
	CRGA	0.017	0.107	0.021	0.128	0.022	0.191	0.030	0.299
	RNC	0.055	0.424	0.050	0.407	0.051	0.503	0.052	**0.561**
	StockCL	**0.057**	**0.441**	**0.052**	**0.433**	**0.056**	**0.563**	**0.053**	0.552

Comparison Methods. We compare our proposed StockCL with the original stock trend forecasting models and several existing contrastive regression methods. ConAlign [1] introduces a contrastive label-aware InfoNCE loss, incorporating conditional alignment and global uniformity. CRGA [16] proposes a contrastive regression method considering label similarity. RNC [20] introduces a label rank based contrastive method.

4.2 Performance comparison

Tab. 2 provides a summary of the performance of StockCL and three baselines on the CSI300 and CSI500 datasets and four stock trend forecasting models. ConAlign and RNC are strong baselines in certain cases. Despite not specifically tailored for stock trend forecasting, ConAlign and RNC showcase the utility of supervised contrastive learning in this regression task. CRGA is the worst baseline, probably because the heuristic similarity function in CRGA is not suitable for stock trend forecasting tasks. In the meanwhile, the RNC result on Localformer is excluded due to the $O(b^3)$ complexity, which leads to an OOM with the large batch size in Localformer. The complexity of RNC limits its application. StockCL surpasses on the four stock trend forecasting models by $6\%, 18\%, 5\%$, 11% on the two datasets in average and generally outperforms the three baseline methods. This highlights the effectiveness of our StockCL framework.

Table 3. Results of ablation study on CSI300 and CSI500 dataset using ALSTM [12] as stock trend forecasting model.

	CSI300				CSI500				average
	IC	ICIR	RankIC	RankICIR	IC	ICIR	RankIC	RankICIR	perf.drop
base	0.053	0.365	0.053	0.362	0.052	0.533	0.056	0.620	-14%
w/o.cp	0.059	0.468	0.055	0.450	0.055	0.543	0.050	0.528	-10%
w/o.y	0.061	0.482	0.056	0.441	0.057	0.558	0.057	0.584	-6%
w/o.neg	0.064	0.512	0.056	0.474	0.057	0.570	0.055	0.604	-4%
StockCL	**0.066**	**0.546**	**0.056**	**0.502**	**0.059**	**0.604**	**0.057**	**0.628**	-

4.3 Ablation study

To evaluate the effectiveness of each design in StockCL, we conduct ablation study to compare StockCL with some variants, as shown in Tab. 3. ***w/o cp*** removes the concept pool component in StockCL and select positive and negative pairs based on label similarity. The average 10% performance drop demonstrates the importance of learning concepts and filtering out the noisy positive pairs defined by label similarity. ***w/o y*** removes the selection of positive pairs by labels in concept-aware contrastive pair selector component. A 6% decrease shows that though concept pool component can detect sample pairs that share the same concept and taking all samples sharing the same concept as positive pairs is sub-optimal. ***w/o neg*** removes the selection of negative pairs by labels in concept-aware contrastive pair selector component. A 4% drop shows the importance of selecting negative samples. Our StockCL framework outperforms all variants in terms of IC, ICIR, RankIC, RankICIR, indicating each designed component truly contributes to performance improvement.

4.4 Impact of hyper-parameter settings

Semantically, there are some important hyper-parameters, the concept number M, the two thresholds ϵ on label similarity and the weight of contrastive learning loss L_{con}. The results on hyper-parameters are shown in Fig.2. For the concept number M, the model performance improves as M increases, peaks at 256 and then slightly decreases. We think before 256, there are actually more concepts and selecting contrastive pair according to a smaller concept set is sub-optimal. When M is greater than 256, there are too many concepts to be learned and some valuable contrastive pairs are filtered. In general, the model performance is insensitive to M in a large scale (128 to 512). For the label similarity threshold ϵ, we maintain $\epsilon_{neg} = \epsilon_{pos} + 0.1$ for simplicity. The performance remains stable from 0.05 to 0.2 and decreases on both sides. This observation aligns with the notion that having too few positive pairs offers insufficient additional supervision, whereas overly broad positive pairs introduce significant noise. For the contrastive loss weight λ, the model performance improves as λ increases, peaks at 0.2 and then apparently decreases. L_{con} is for better representations and do not explicitly reflect the labels. Hence, we need to balance the strength of L_{con}.

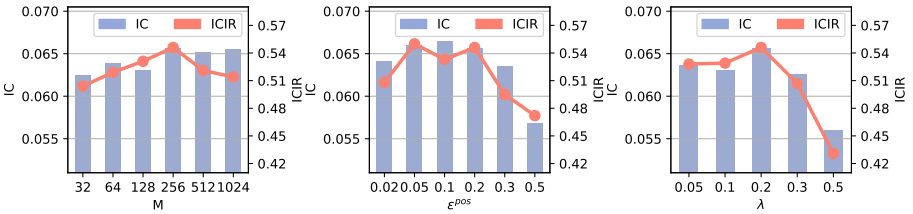

Fig. 2. Impact of hyper-parameters on CSI300 dataset using ALSTM [12] as stock trend forecasting model.

5 Related Works

Stock trend forecasting. Deep learning based stock trend forecasting [2, 5, 10] has been a hot topic in recent years, while most works depend on limited historical stock data for supervised learning, making the model susceptible to overfitting [14]. Some studies [4, 17] turn to minute-level stock data to boost daily stock trend forecasting, where a contrastive learning loss is devised to learn mutual information between minutely trading data and daily data. Our work is orthogonal to them with no constraints on the kind of data.

Contrastive learning. In the general field of deep learning, self-supervised contrastive learning [19] generates contrastive pairs by data augmentation, while supervised contrastive learning [8, 9] groups positive samples according to class labels. As samples with exactly the same label are scarce in regression tasks, recent works [1, 16, 20] heuristically estimate the degree of positivity based on label proximity. In this paper, we offer a novel perspective by learning concepts in stock data to filter out noisy positive samples.

6 Conclusion

In this paper, we propose StockCL to augment supervision signals for stock trend forecasting models. StockCL involves two core components, concept pool and concept-aware contrastive pair selector. The experimental results on two datasets demonstrate that StockCL consistently improves the performance from 5% to 18% for four stock trend forecasting models.

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